

DIVE

A Project Management Framework for Artificial Intelligence Initiatives

Diagnose >> Iterate >> Validate >> Evolve

Vahid Hakami
May 2026 | Version 1.0

Abstract

Artificial Intelligence (AI) projects face fundamental challenges that traditional project management methodologies (PMBOK, PRINCE2, Agile, SAE) were not designed to address: non-deterministic outputs, data-dependent feasibility, model drift, compute cost uncertainty, and ethical risk. Existing approaches assume deterministic outcomes and known solution paths, leading to high failure rates in AI initiatives. This paper introduces the DIVE Framework (Diagnose >> Iterate >> Validate >> Evolve), a purpose-built methodology for managing AI and machine learning projects. DIVE introduces seven novel principles including Metric Contracts, parallel experimentation tracks, feasibility gates, and continuous governance as a lifecycle phase. The framework includes 14 templates and 5 governance artifacts. We present the complete methodology and discuss its differentiation from existing approaches.

1. Introduction

The Project Management Institute reports that over 70% of AI projects fail to deliver expected value [1]. Root cause analyses consistently point to the same factors: unclear success criteria, insufficient data quality, unrealistic timelines, and failure to account for model degradation post-deployment. These are not failures of execution -- they are failures of methodology.

Traditional project management frameworks were designed for deterministic environments. A software feature either works or it does not; requirements can be specified upfront; quality is binary. AI projects operate in fundamentally different conditions:

Characteristic	Traditional Software	AI Projects
Output	Deterministic	Probabilistic
Spec	Known upfront	Discovered through data
Failure mode	Bug -> fix	Data/model degradation
Estimation	Possible with history	High uncertainty
Quality check	Pass/fail tests	Metric thresholds
Main cost	Engineering hours	Compute + data + talent

The DIVE Framework was designed from first principles to address these differences.

2. The Seven Principles

Principle 1: Data First, Code Second

No AI project begins without a data feasibility audit. If the data does not exist, is inaccessible, or lacks signal, the project cannot succeed regardless of model sophistication.

Principle 2: Parallel Exploration Beats Sequential Certainty

AI development is research-like; the optimal approach is unknown at project start. DIVE mandates 3-5 parallel experimentation tracks with fixed compute budgets.

Principle 3: Metric Contracts Replace Requirements Documents

Success criteria are defined as measurable, signed contracts (e.g., "F1 $\geq$ 0.87, latency $\leq$ 200ms p95") before development begins. These are the sole criteria for phase exit.

Principle 4: Fail Fast, Fail Quantitatively

Pre-defined quantitative kill criteria prevent indefinite "one more iteration" cycles. A kill decision is a success -- it preserves resources.

Principle 5: Governance Is Continuous, Not Periodic

Post-deployment drift (data drift, concept drift, performance drift) means models degrade even if code does not change. Continuous monitoring and governance are a permanent lifecycle phase.

Principle 6: Human-in-the-Loop Is a Design Constraint

Three escalation tiers (Advisory, Assisted, Automated) are designed pre-deployment, with confidence thresholds and fallback routing.

Principle 7: Compute Is a First-Class Resource

GPU time, API tokens, and storage costs are tracked alongside scope, schedule, and budget. Each experiment track has a fixed compute budget.

3. The Lifecycle

The DIVE lifecycle consists of four phases:

Diagnose >> Iterate >> Validate >> Evolve

Phase 1 -- Diagnose (2-4 weeks)

Problem validation, data audit, feasibility baseline, Metric Contract negotiation, ethics screening, risk registration, build/buy assessment. Exit gate: Feasibility Review (Go/No-Go/Redesign).

Phase 2 -- Iterate (4-8 weeks)

3-5 parallel experiment tracks with fixed compute budgets. Weekly comparison on a shared dashboard. Compute burn rate tracking. Mid-phase pivot allowed. Exit gate: Candidate Selection.

Phase 3 -- Validate (2-4 weeks)

Hold-out evaluation, robustness testing, bias audit, performance benchmarking, shadow deployment (optional), stakeholder sign-off. Exit gate: Validation Sign-Off.

Phase 4 -- Evolve (ongoing)

Production deployment, monitoring configuration, drift detection, feedback pipeline, quarterly governance reviews, retraining as needed. Exit gate: Decommissioning.

4. Differentiation from Existing Approaches

Dimension	PMBOK	PRINCE2	Agile	DIVE
AI-native	No	No	No	Yes
Data feasibility gate	No	No	No	Yes
Parallel exploration	No	No	No	Yes
Metric Contracts	No	No	No	Yes
Compute as resource	No	No	No	Yes
Continuous governance	No	No	No	Yes
Explicit kill criteria	No	No	No	Yes

5. Conclusion

The DIVE Framework provides a structured, AI-native approach to project management that addresses the unique challenges of data-driven, non-deterministic, continuously evolving AI systems. By replacing false certainty with measurable gates, parallel exploration, and continuous governance, DIVE aims to improve the success rate of AI initiatives across industries. The framework is vendor-agnostic, tool-agnostic, and designed to scale from small teams to enterprise AI programs.

References

[1] Project Management Institute. "AI Project Success Rates and Root Causes." PMI White Paper, 2024.